

LEDGER · 2026-08-28 · INDUSTRIAL BIOTECHNOLOGY / POLYMER SCIENCE

Premium High-Barrier Packaging Plastics (PHBV) via Non-Sterile Halophilic Fermentation

REJECTED

This concept did **not** clear the framework.
The reasoning is published in full below.
What the assessment found

Incumbent benchmark	Polyethylene (PE)
Measured advantage	PHBV OTR of 218.4 cm ³ /m ² /day vs Polyethylene OTR of 1543.09 cm ³ /m ² /day (7.0x lower oxygen permeability).
Time to first revenue	24 months
Regulatory risk	Moderate — qualification costs reported (\$1,500,000 estimated)

Why it failed

- Voided by: Mismatched comparison

Assessment

Polyethylene is specified by buyers for moisture barrier and heat sealing, not oxygen barrier performance, making the OTR baseline comparison non-like-for-like. Additionally, FDA food-contact notifications and packaging converter qualification cycles will extend time to first revenue beyond 18 months.

Also flagged

- Evidence rests on non-peer-reviewed sources
- no_production_runsheet

A rejection is not a claim that the science is wrong. It means the venture case did not survive: the comparator was not what the buyer actually buys, the comparison was not like-for-like, the advantage fell short of a step change, or the capital and time to first revenue were prohibitive.

Assessed 2026-08-28 against seven gates plus the voiding sub-tests. [How the method works](#) · [Browse all concepts](#)

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Method

Ledger

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Track record

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